

Chapter 10: WEB DEVELOPMENT

Skills and Functions Needed for a Successful Large-Scale Project

Project Manager Information Architect

User Experience Designer Marketing Representative Copy

Writer & Editor Content Manager

Graphic Designer Database Administrator Network

Administrator

Web Developer / Web Designer

Skills and Functions Needed for a Successful Small Project

The skills and functions are essentially the same as on a large project

Each person may wear many “hats” and juggle their job roles

- Example: The web designer may also be the graphic designer.

Some job roles may be outsourced

- Most Common:

An external web site provider is used so there is less (if any) need for a Network Administrator.

Project Staffing Criteria

- Finding the right people to work on a project is crucial
 - Consider:
 - work experience
 - portfolio
 - formal education

- industry certifications

- Alternative: Outsource all or portions of the project to a web design/development firm

Project Job Roles

- Project Manager
- Information Architect
- UX Designer
- Marketing Representative
- Copywriter & Editor
- Content Manager
- Graphic Designer
- Web Designer
- Web Developer
- Front-End Web Developer
- Back-End Web Developer
- Database Administrator
- Network Administrator

Website Development Cycle

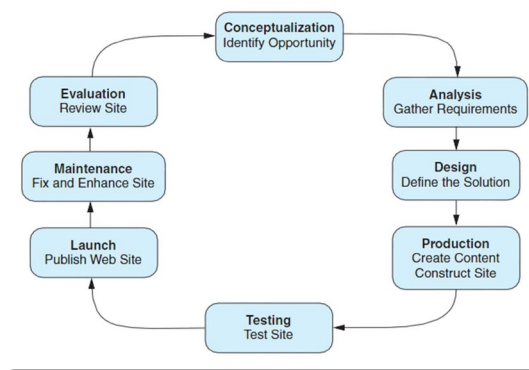


Figure 10.2 The Website Development Cycle

Web Development: Conceptualization

- Determine the intended audience
- Determine the goals or mission of the web site
 - Short-term goals
 - Long-term goals
 - Develop measurable objectives such as:
 - Number of visitors
 - Percent of product sales

Web Development: Analysis

- Determine the following:
 - information topics
 - functionality requirements (high-level)
 - Determine “what” a site will do – not “how” it will do it
 - environmental requirements
 - content requirements
- If a redesign, compare the old approach with the new approach
- Review competitor’s sites
- Estimate costs
 - Cost/Benefit analysis

Web Development: Design

- Determine the site organization
- Prototype the design
- Determine a page layout design
- Document each page

Wireframe

A sketch or blueprint of a web page

Shows the structure of the basic page elements, including:

- Header
- Navigation
- Content
- Footer
- Image locations

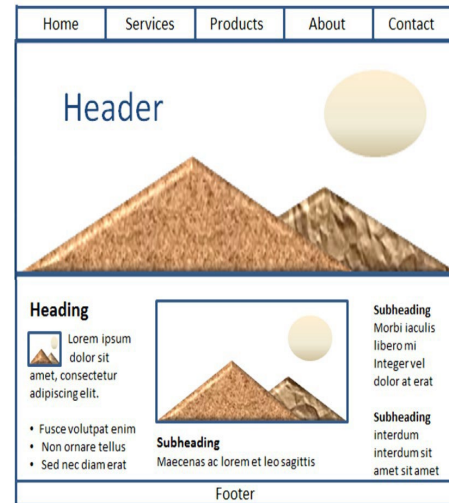


Figure 5.35 This wireframe page layout a top navigation area and a hero image

Web Development: Production

- Choose a web authoring tool
- Organize your site files
- Develop and individually test components
- Add content

Web Development: Testing

- Test on different web browsers and browser versions
- Test with different screen resolutions
- Test using different bandwidths
- Test from another location
- Test, Test, Test

Web Development: Types of Testing

Automated Testing Tools and Validation

- Automated Testing (Link checkers, etc.)
- W3C HTML and CSS validation tests

Usability Testing

- Testing how actual web page visitors use a web site
- Can be done at almost any stage of development
 - Early – use paper and sketches of pages
 - Design – use prototype
 - Production & Testing – use actual pages

Web Development: Approval & Launch

User or Client Testing

- Client will test site before giving official approval for site launch

Approval & Launch

- Obtain sign-off form or email from client
- Upload files to web server
- Create backup copies of files
- TEST THE WEB SITE AGAIN AFTER UPLOAD!

Web Development: Maintenance

Maintenance – the never-ending task...

- Enhancements to site
- Fixes to site
- New areas added to site

A new opportunity or issue is identified and another loop through the development process begins.

Web Development: Evaluation

- Re-visit the goals, objectives, and mission of the website
- Determine how closely they are being met
- Develop a plan to better meet the goals, objectives and mission

File Organization

Relative links from the canyon.html page in the rooms folder

```
<a href="javelina.html">Javelina</a>
```

```
<a href="../index.html">Home</a>
```

```
<a href="../events/weekend.html">Weekend Events</a>
```

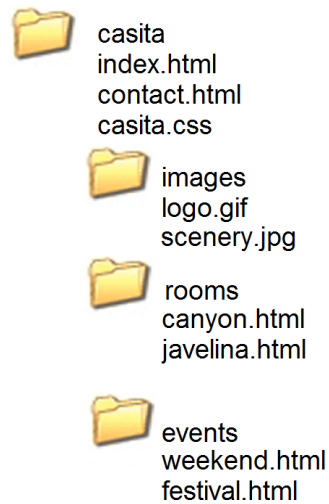



Figure 10.5 The web page files are organized in folders.

Domain Names

Establishes a web presence for your business or organization

- New Business
 - choose domain name while selecting company name
- Established Business
 - choose a domain name that relates to your established business presence

Choosing a Domain Name

- Describe your business
- Be brief, if possible
- Avoid hyphens

- TLD (top level domain name)
 - .com, .net, .biz, .us, .mobi and others
 - .org for non-profits
- Brainstorm potential keywords
- Avoid trademarked words or phrases
- Know the territory (use Google!)
- Verify Availability

Registering a Domain Name

- There are many domain name registrars, including
 - <https://register.com>
 - <https://networksolutions.com>
 - <https://godaddy.com>
- Visit Registrar, choose name, pay
- Private registration – additional fee but personal information is kept private and out of the WHOIS database

Web Hosting

Some organizations administer a web server in-house and host their own website.

Many companies use a web host provider.

- A good web hosting service will provide a robust, reliable home for your website.
- A poor web hosting service will be the source of problems and complaints.
- Do not use free web hosting for a commercial website.

Types of Web Hosting

Hosting Needs: Small to Medium Website

- Virtual Hosting
 - The web host provider's server is divided into a number of virtual domains and multiple websites are set up on the same computer.
- Consider:
 - Future growth and scalability of web host
 - Operating System
 - Types of server-side processing supported
 - Bandwidth of Internet connection
 - Both local and national web host providers
 - Guaranteed uptime – service level agreement (SLA)
 - Technical support

Hosting Needs: Large to Enterprise Website

- Dedicated web Server
- Co-located web Server
- Consider:
 - National web host providers
 - Guaranteed uptime – service level agreement (SLA)
 - Bandwidth of Internet connection
 - Technical Support

Dedicated Web Server

Dedicated Web Server

- The exclusive use of a rented computer and connection to the Internet that is housed in the web hosting company's premises.
- The server can usually be configured and operated remotely from the client company or you can pay the web host provider to administer it for you.

Co-Located Web Server

Co-Located Web Server

- A computer that your organization has purchased and configured.
- Your organization effectively rents space at the web host provider's location.
- Your server is kept and connected to the Internet at their location.
- Your organization administers this computer.

Choosing a Virtual Host

See the Web Host Checklist (Table 10.1)

- Operating System
- Web Server Software
- Bandwidth
- Technical Support
- Service Agreement
- Disk Space
- E-mail
- Uploading Files
- Canned Scripts

- Scripting Support
- Database Support
- E-Commerce Packages
- SSL Encryption
- Scalability
- Backups
- Site Statistics
- Domain Name
- Price

Summary

- This chapter introduced you to the system development life cycle and its application to web development projects.
- The job roles related to website development were also discussed.
- An introduction to selecting a domain name and a web host was provided.